1. **Define AI**

**Artificial intelligence is concern with the design of intelligence in artificial systems**

The term Artificial intelligence (AI) is coined by **John McCarthy.**

- Artificial intelligence is the science of making machines that can think like humans.
- The design and study of computer programs that behave intelligently.
- Artificial intelligence (AI) is a wide-ranging branch of computer science concerned with building smart machines capable of performing tasks that typically require human intelligence.
- Artificial intelligence (AI) is a set of technologies that enable computers to perform a variety of advanced functions, including the ability to see, understand and translate spoken and written language, analyze data, make recommendations, and more.

2. **Some of the most common examples of AI in use today include:**

**ChatGPT:** Uses large language models (LLMs) to generate text in response to questions or comments posed to it.

**Google Translate:** Uses deep learning models to translate text from one language to another.

**Netflix:** Uses machine learning to create personalized recommendations for users based on their previous viewing history.

**Tesla:** Uses computer vision to power self-driving features on their cars.

3. **Define AI with respect to thinking humanly**

   The branch of science and technology that is devoted to the creation of machines that learn and think as intelligently as human beings is known as AI

4. **Define AI with respect to thinking rationally**.

It is the ability to make decisions based on logical reasoning and optimize behaviour to achieve its goals, considering its perception of the environment and the performance measure.

5. **Define AI with respect to acting humanly**.

It means attempting to model and understand how human mind works.

6. **Define AI with respect to acting rationally.**

   Acting rationally means acting so as to achieve one's goals, given one's beliefs. An agent is something that perceives and acts**.**

7. **What is intelligence and Knowledge?**

Knowledge is the collection of skills and information a person has acquired through experience. Knowledge refers to the fact of knowing something that has been gained through an experience and learning which makes the agent familiar.

Intelligence is the ability to apply knowledge. Intelligence is the capacity to learn, to grasp new ideas, and to accurately see relationships between things.

**8. What is Turing test?**

The Turing Test is a method of inquiry in artificial intelligence (AI) for determining whether or not a computer is capable of thinking like a human being. The test is named after Alan Turing, the founder of the Turing Test and an English computer scientist, cryptanalyst, mathematician and theoretical biologist.

List the capabilities needed by computers to pass Turing test.

**9. Who is the father of AI and Who coined the term Artificial intelligence?**

The term Artificial intelligence (AI) is coined by **John McCarthy** and he is known as father of AI.

**10. What is intelligence?**

Intelligence is the ability to apply knowledge. Intelligence is the capacity to learn, to grasp new ideas, and to accurately see relationships between things. Intelligence is ability to think and act rationally.

**11. Mention any 2 domains where human beings are better than Computers.**

- Common sense/Judgment based reasoning.
- Langauage,speech
- Sense the situations, understanding.

**12. Mention any 2 domains where computers are better than human beings.**

- Numeric Computations
- Information storage
- Repetitive operations
-

**13. Define Cognitive Science and Linguistics.**

Cognitive science is an investigatory discipline that explores intelligence and the human mind. **Cognitive science is the study** of the human mind and brain, focusing on how the mind represents and manipulates knowledge and how mental representations and processes are realized in the brain.
**Linguistics** is the scientific study of language, and its focus is the systematic investigation of the properties of particular languages as well as the characteristics of language in general.

**14. What is Speech Recognition?**

**Speech recognition or speech-to-text recognition**, is the capacity of a machine or program to recognize spoken words and transform them into

text. Speech Recognition is an important feature in several applications used such as home automation, artificial intelligence, etc.

**15. Mention any 4 application areas of AI.**

AI in E-Commerce
AI in Education
AI in Healthcare
AI in Robotics
AI in Automobiles

**16. What is the importance of AI?**

***Artificial Intelligence (AI)*** has evolved rapidly, becoming a transformative technology in many areas of modern life. At its core, AI involves the development of computer systems capable of performing tasks that traditionally require human intelligence, such as visual perception, speech recognition, decision-making, and language translation.

**AI matters for several reasons:**

- Efficiency and Automation
- Insights and Decision Making
- Personalization
- Addressing Complex Problems
- Competitive Advantage
- Innovation
- Cost Savings

**17. What are the related fields in AI?**

- Robotics
- Machine Learning
- Expert System
- Neural Network
- Data Science
- Natural Language Processing

**18. Who created the first Programming language for AI research? What is the name of that Programming language?**

The first practical and still most widely used AI programming language is the functional language **Lisp developed by John McCarthy** in the late 1950s. Lisp is based on mathematical function theory and the lambda abstraction.

**19. Expand AAAI and LISP**

AAAI- Association for the Advancement of Artificial Intelligence (AAAI). It is the premier scientific society

**LISP - LISt Processing.**

The Lisp language is the oldest and most widely used functional language. Lisp has long been a popular language for applications in artificial intelligence.

20. What is Conversational AI? Give any two examples for conversational AI

    Conversational AI refers to the ability of computers to understand intent and mimic human emotions to simulate natural conversations in response to text or speech inputs.

    **Examples:**

    - Chatbots
    - Voice assistants
    - Siri
    - Amazon Alexa.

21. List any 4 state of the art AI applications.

    - Autonomous planning and scheduling
    - Game playing
    - Autonomous control
    - Diagnosis
    - Rootics
    - Logistics planning

**22. What is an agent? Give example. How agent performs the actions?**

An "agent" is anything that can be viewed **as perceiving its environment through sensors** and acting upon that environment **through actuators.**

Example: robotic agent, software agent, human agent

**Sensor** is a device which detects the change in the environment and sends the information to other electronic devices. **Actuators** are the component of machines that converts energy into motion. An actuator can be an electric motor, gears, rails, etc.

**Effectors** are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

AI agents are systems that can perform tasks autonomously by interpreting data from the environment, making decisions based on that data, and executing actions to achieve set goals. They operate under predefined algorithms and machine learning models that enable them to learn and adapt over time.

**23. List various terminologies associated with agent and environment**

**Sensor** is a device which detects the change in the environment and sends the information to other electronic devices.

**Actuators** are the component of machines that converts energy into motion. An actuator can be an electric motor, gears, rails, etc.

**Effectors** are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

24. **Define percept and percept sequence.**

    Percept refers to the agent's perceptual inputs at any given instant.

    Percept sequence is the complete history of everything the agent has ever perceived.

**25. Define agent function and agent program.**

Agent's behavior is described by agent function that maps any given percept sequence to an action.
It is an abstract mathematical description.

**Agent program** is a concrete implementation, running on the agent architecture. It is a implementation of agent function.

26. **Define rational agent.**

It is one that does the right thing.
***For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built in knowledge the agent has.***

27. **List various Agent Terminologies**

**Performance Measure of Agent** − It is the criteria, which determines how successful an agent is.
**Behavior of Agent** − It is the action that agent performs after any given sequence of percepts.
**Percept** − It is agent's perceptual inputs at a given instance.
**Percept Sequence** − It is the history of all that an agent has perceived till date.
**Agent Function** − It is a map from the precept sequence to an action.

**28. Differentiate omniscience and rationality.**

An omniscience agent knows the actual outcome of its actions and can act accordingly.
Rationality maximizes expected performance of an agent. Perfection maximizes actual performance.

**29. What is an intelligent agent? What are the 2 main features of an intelligent agent?**

An intelligent agent is a program that can make decisions or perform a service based on its environment, user input and experiences.
These programs can be used to autonomously gather information on a regular, programmed schedule or when prompted by the user in real time.

- Intelligent agents in AI **are autonomous entities** that act upon an environment using sensors and actuators to achieve their goals.
- Intelligent agents **may learn from the environment to achieve those goals**.
- Intelligent agents are commonly used to solve complex tasks that are difficult or impossible for humans to do

Driverless cars and the Siri virtual assistant are examples of intelligent agents in AI

**30. What are Actuators and Effectors?**

An **Effector is any device** that affects the physical environment. **An actuator is the actual mechanism** that enables the **effector to execute** an action.

**Actuators:** Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.
**Effectors:** Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

**31. What is an intelligent agent? Give examples for Intelligent agents in AI**
An intelligent agent is a program that can make decisions or perform a service based on its environment, user input and experiences.

Driverless cars and the Siri virtual assistant are examples of intelligent agents in AI.

**32. What is an environment in AI ?**
The surroundings or circumstances in which an AI system functions are referred to as the environment in AI. It includes the physical environment, digital platforms, and virtualized worlds where AI models and algorithms are used.

**33. What is task Environment? Give example.**
The first step in designing an AI agent is to specify the task environment. **The task environment is comprised of PEAS (Performance measure, Environment, Actuators, Sensors).**
Vacuum world is a simple example. another example is automated taxi driver.

**34. Define performance measure.**
It is the criteria, which determines how successful an agent is.

**35. Write PEAS for automated taxi driver.**
Agent type –Automated taxi driver
**Performance measure-** safe, fast, legal. Comfortable trip, Maximize profit, minimum cost
**Environment-** Road, traffic, pedestrians, customers.
**Actuators-** Steering, brake accelerator, signal, horn
**Sensors-** camera, GPS, odometer, engine sensor, keyboard

**36. What do you mean by fully observable task environment? Give example.**

If an agent's sensors give it access to the complete state of the environment at each point in time, then we say that the task environment is fully observable.
A task environment is effectively fully observable if the sensors detect all aspects that are relevant to the choice of action, relevance depends on the performance measure.
Example: Online shopping, Playing Chess, Driving a car

**37. What is Agent Function and agent Program?**

**Agent Function** − It is a map from the precept sequence to an action or It is a mathematical function that maps a sequence of perceptions into action.

**Agent Program-** It is an implementation of an agent function.

**38. List the types of Agents.**

- Simple reflex agents.
- Model-based agents.
- Goal-based agents.
- Utility-based agents.
- Learning agents.
- Hierarchical agents.

**39. What are goal based agents? Give example**

Goal-based virtual agents in AI are designed to achieve specific objectives. They possess a goal formulation capability, allowing them to decide what goals to pursue and prioritize.

**Example**: Self-driving cars

**40. List the four basic kinds of agents programs.**

1. Simple reflex agents
2. Model based reflex agents.
3. Goal based agents
4. Utility based agents.

**41. Limitations of table driven agent.**

1. Size of the table
2. Time to build the table.
3. No autonomy
4. Time required for learning.

**42. What is simple reflex agent? Give example.**

Simple reflex agents are the most basic types of intelligent agents in AI. They operate based on a set of predefined rules or condition-action pairs. These agents make decisions solely based on the current percept, which is the information they gather from their environment at a given moment

**Example:** A thermostat is a simple reflex agent that regulates room temperature based on a predefined set point.

**43. Limitations of simple reflex agent.**

They cannot reason or plan and are often employed in situations with straightforward, predictable outcomes.

**44. What are model based agents? Give example.**

Model-based reflex agents are an improvement over simple reflex agents. They maintain an internal model or representation of the world. This model allows them to consider the history of their interactions with the environment and make more informed decisions.

**Example**: Chess-playing programs like IBM's Deep Blue use a model-based approach.

**45. What are utility based agents**

Utility-based agents take goal-based AI a step further by considering the desirability or utility of different outcomes. They assign values or utilities to various states or situations and make decisions that maximize expected utility.

**Self-driving cars use utility-based agents** to make driving decisions.

**46. What is Utility function?**

Utility function maps a state onto a real number, which describes the associated degree of happiness.

**47. List the four conceptual components of a learning agent.**

1. Learning element
2. Performance element.
3. Critic.
4. Problem generator.

48. Explain learning agent.

**Learning element-** responsible for making improvements.
**Performance element-** responsible for selecting external actions.
**Critic.-** it is used for feedback, how the agent is doing and determines how the performance element should be modified to do better in the future.
**Problem generator-** It is responsible for suggesting actions that will lead to new informative experiences.

**49. What are the future prospects and challenges in AI?**

In the coming decades, we can expect the following key changes in AI:

**1. Self-learning**: Artificial intelligence will become increasingly capable of self-learning, allowing systems to learn on the fly without requiring constant reprogramming.

**2. Global AI Network**: When AI begins to share data and knowledge, we can expect to see the development of a global AI network, which will open up new perspectives for solving the world's complex problems.
**3. Ethics and Safety**: Along with AI capabilities come ethical and safety issues. The development of standards and regulations in this area is becoming an integral part of the future of AI.

**50. Explain Rationality. (Long answer)**

Rationality at any given time depends on

- The performance measure that defines the criterion of success
- The agent's prior knowledge
- The actions that the agent can perform
- The agent's percept sequence to date

**This leads to definition of rational agents as:**

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built in knowledge the agent has.

Example: Simple vacuum cleaner agent that cleans a square if it is dirty and moves to other square if not;

Let us assume-

- Performance measure awards one point for each clean square at each time
- The geography of the environment is known. But dirt distribution and initial location of agent are not known.
- The available actions are Left, Right, Suck and NoOp
- The agent correctly perceives its location.

We say that under these circumstances the agent is rational. But one can easily see that the same agent would be irrational under different circumstances. So we need to differentiate between rationality and omniscience.

51. Explain "Acting humanly, **the turing test approach**.

It was proposed by the British mathematician and computer scientist Alan Turing in 1950 in his seminal paper, "Computing Machinery and Intelligence." He proposed that the **"Turing test is used to determine whether or not a computer(machine) can think intelligently like humans"?**

It is a test to see if a computer can learn to mimic human behavior. He defined intelligent behavior as the ability to achieve human-level intelligence during a conversation. This performance should be sufficient to trick an interrogator into thinking that the answers are coming from a human.

To see if a machine can do this, he proposed a test setup: he proposed that a human should interrogate the machine through a text interface. Another constraint is that the human cannot know who's on the other side of the interrogation, which means it can either be a machine or a human. To

enable this setup, a human will be interacting with two entities through a text interface. These two entities are called respondents. One of them will be a human and the other one will be the machine.
The respondent machine passes the test if the interrogator is unable to tell whether the answers are coming from a machine or a human.

At the very minimum, the machine needs to be well versed with the following things:
**Natural Language Processing:** The machine needs this to communicate with the interrogator. The machine needs to parse the sentence, extract the context, and give an appropriate answer.
**Knowledge Representation:** The machine needs to store the information provided before the interrogation. It also needs to keep track of the information being provided during the conversation so that it can respond appropriately if it comes up again.
**Reasoning:** It's important for the machine to understand how to interpret the information that gets stored. Humans tend to do this automatically to draw conclusions in real time.
Machine Learning: This is needed so that the machine can adapt to new conditions in real time. The machine needs to analyze and detect patterns so that it can draw inferences.

## *52. Thinking humanly: The cognitive modeling approach*

Once we gather enough data, we can create a model to simulate the human process. This model can be used to create software that can think like humans. Of course this is easier said than done! All we care about is the output of the program given a particular input. If the program behaves in a way that matches human behavior, then we can say that humans have a similar thinking mechanism.
Within computer science, there is a field of study called **Cognitive Modeling** that deals with simulating the human thinking process. It tries to understand how humans solve problems. It takes the mental processes that go into this problem solving process and turns it into a software model. This model can then be used to simulate human behavior. Cognitive modeling is used in a variety of AI applications such as deep learning, expert systems, Natural Language Processing, robotics, and so on.
The interdisciplinary field of cognitive science brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.

## 53. Thinking rationally: The "laws of thought" approach

Rationality refers to doing the right thing in a given circumstance.
The Greek philosopher Aristotle was one of the first to attempt to codify "right thinking," that is, irrefutable reasoning processes. His **syllogisms** provided patterns for argument structures that always yielded correct conclusions when given correct premises — for example, *"Socrates is a man; all men are mortal; therefore, Socrates is mortal."* These laws of thought were supposed to govern the operation of the mind; their study initiated the field called **logic.**
***There are two main obstacles to this approach.***

**1.** It is not easy to take informal knowledge and state it in the formal terms required by logical notation, particularly when the knowledge is less than 100% certain.

**2.** There is a big difference between solving a problem "in principle" and solving it in practice. Even problems with just a few hundred facts can exhaust the computational resources of any computer unless it has some guidance as to which reasoning steps to try first.

**54.Acting rationally: The rational agent approach**

An agent is just something that acts (agent comes from the Latin agere, to do). Of course, all computer programs do something, but computer agents are expected to do more: operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals.

Rational agents need to be performed in such a way that there is maximum benefit to the entity performing the action. An agent is said to act rationally if, given a set of rules, it takes actions to achieve its goals. It just perceives and acts according to the information that's available. This system is used a lot in AI to design robots when they are sent to navigate unknown terrains.

All the skills needed for the Turing Test also allow an agent to act rationally. Knowledge representation and reasoning enable agents to reach good decisions. We need to be able to generate comprehensible sentences in natural language to get by in a complex society.

***The rational-agent approach has two advantages over the other approaches.***

**1.** It is more general than the "laws of thought" approach because correct inference is just one of several possible mechanisms for achieving rationality.

**2.** It is more amenable to scientific development than are approaches based on human behavior or human thought.